

Incident Response Report

PREPARED BY
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DATE ISSUED
30 June 2026

Incident Information

Field	Value
Incident ID	IR-2026-0630-0345
Incident Name	POST-EXPLOITATION ATTACK CHAIN (PowerShell Execution → Registry Persistence → Privilege Escalation)
Analyst	Othmane Benmezian
Date	30 June 2026
Affected Host	Endpoint Windows 10 Pro
Detection Platform	Wazuh SIEM + Sysmon
Severity	High
Status	Closed

1 - Executive Summary

During routine security monitoring, a high-severity alert was generated for suspicious PowerShell activity on the Windows 10 Pro endpoint assigned to Amin. The alert was triggered by a custom Wazuh detection rule designed to identify potentially malicious PowerShell execution.

Initial analysis confirmed that the recorded event matched the expected behavior defined by the custom detection rule. The activity suggested that an attacker, after obtaining access to the endpoint, executed PowerShell with the apparent objective of preparing the system for subsequent malicious actions while minimizing traces of execution.

Shortly after the initial execution, additional alerts indicated the creation of a Registry Run Key, establishing persistence on the compromised host. Further investigation revealed another alert showing that the same user account had been added to the local Administrators group, indicating a successful privilege escalation attempt. All three attack stages occurred within a very short timeframe, confirming a coordinated multi-stage attack.

The investigation determined that the incident was limited to the Windows 10 Pro endpoint assigned to Amin, with no evidence of lateral movement to other systems. Immediate containment and remediation actions were performed to remove persistence, revoke the unauthorized administrative privileges, verify endpoint integrity, and restore the affected system to a secure operational state.

2. Incident Overview

2.1 Lab Environment

System	Role
 Windows 10 Pro	Target endpoint monitored by Wazuh Agent
 Windows Server 2022 (Datacenter)	Active Directory Domain Controller
 Ubuntu Server	SIEM server running Wazuh Manager (Docker)
 Kali Linux	Attacker machine used for controlled attack simulation (Hydra)

2.2 Attack Scenario

This incident simulates a multi-stage attack consisting of:

1. Suspicious PowerShell Execution
2. Registry Run Key Persistence
3. Privilege Escalation by adding a user to the Administrators group

3. Attack Timeline

Time (UTC)	Event
13:54:29	Suspicious PowerShell executed (powershell.exe -ExecutionPolicy Bypass -NoProfile)
13:54:32	Wazuh generated PowerShell Execution alert (Rule 100201)
13:54:46	Registry Run Key (HKCU\Software\Microsoft\Windows\CurrentVersion\Run\Updater) created
13:54:49	Wazuh generated Registry Persistence alert (Rule 92302)
13:54:55	User testuser added to the local Administrators group (Event ID 4732)
13:54:58	Wazuh generated Privilege Escalation alert (Rule 60154)
~13:56	Analyst initiated containment and remediation actions
~13:58	Incident investigation completed and incident closed

4. Detection

4.1 Suspicious PowerShell Execution

# input.type	log
# location	EventChannel
# manager.name	wazuh.manager
# rule.description	Suspicious PowerShell execution detected
# rule.firedtimes	2
# rule.groups	lab, windows, powershell, execution, powershell, lab
# rule.id	100201
# rule.level	10
rule.mail	false
# rule.mitre.id	T1059.001
# rule.mitre.tactic	Execution
# rule.mitre.technique	PowerShell
timestamp	Jun 30, 2026 @ 14:54:32.181

Detection Details

Field	Value
Triggering Event	Windows Security Event ID 4688 (Process Creation)
Command Line	powershell.exe -ExecutionPolicy Bypass -NoProfile
Parent Process	powershell.exe
User	SOC estuser
Host	Client.SOC.local (Amin)
Rule ID	100201
MITRE ATT&CK	T1059.001 - PowerShell

4.2 Registry Persistence

# input.type	log
# location	EventChannel
# manager.name	wazuh.manager
# rule.description	Registry entry to be executed on next logon was modified using command line application reg.exe
# rule.firedtimes	2
# rule.groups	sysmon, sysmon_eid13_detections, windows
# rule.id	92302
# rule.level	6
# rule.mail	false
# rule.mitre.id	T1547.001
# rule.mitre.tactic	Persistence, Privilege Escalation
# rule.mitre.technique	Registry Run Keys / Startup Folder
# timestamp	Jun 30, 2026 @ 14:54:49.437

Detection Details

Field	Value
Registry Path	HKCU\Software\Microsoft\Windows\CurrentVersion\Run\Updater
Registry Value	Updater
Value Data	C:\Windows\System32\calc.exe
Process Responsible	C:\Windows\System32\reg.exe
Event ID	Sysmon Event ID 13 (Registry Value Set)
Rule ID	92302
MITRE ATT&CK	T1547.001 - Registry Run Keys / Startup Folder

4.3 Privilege Escalation

# rule.description	Administrators Group Changed
# rule.firedtimes	3
# rule.gdpr	IV_35.7.d, IV_32.2
# rule.gpg13	7.10
# rule.groups	windows, windows_security, group_changed, win_group_changed
# rule.hipaa	164.312.a.2.I, 164.312.a.2.II, 164.312.b
# rule.id	60154
# rule.level	12
# rule.mail	true
# rule.mitre.id	T1484
# rule.mitre.tactic	Defense Evasion, Privilege Escalation
# rule.mitre.technique	Domain Policy Modification
# rule.nist_800_53	AC.2, IA.4, AU.14, AC.7
# rule.pci_dss	8.1.2, 10.2.5
# rule.tsc	CC6.8, CC7.2, CC7.3
# timestamp	Jun 30, 2026 @ 14:54:58.013

Detection Details

Field	Value
User Added	testuser
Target Group	Builtin\Administrators
Event ID	4732 - A member was added to a security-enabled local group
Host	Client.SOC.local (Amin)
Rule ID	60154
MITRE ATT&CK	T1484 - Privilege Escalation

5. Investigation

5.1 PowerShell Investigation

Initial Findings

Field	Value
Executing User	SOC\testuser
Process	powershell.exe
Parent Process	powershell.exe
Command Line	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -ExecutionPolicy Bypass -NoProfile
Timestamp	2026-06-30 13:54:32 UTC

Analysis

The investigation began after Wazuh generated a Suspicious PowerShell Execution alert.

Reviewing the Security Event ID 4688 showed that the PowerShell process was launched using the following arguments:

- ExecutionPolicy Bypass
- NoProfile

The use of ExecutionPolicy Bypass disables PowerShell execution restrictions, which is a well-known technique used by attackers to execute scripts while bypassing local security policies.

The process was executed by SOC\testuser, indicating that the attacker was already operating under a valid user context following the initial compromise. The timing of this event is significant because it immediately preceded registry persistence creation and privilege escalation activities, suggesting it was used as the execution stage of the attack chain.

No additional malicious PowerShell child processes were observed during the investigation.

Indicators of Compromise (IOCs)

- powershell.exe
- ExecutionPolicy Bypass
- NoProfile
- Security Event ID 4688

5.2 Registry Investigation

Initial Findings

Field	Value
Registry Path	HKCU\Software\Microsoft\Windows\CurrentVersion\Run
Registry Value	Updater
Value Data	C:\Windows\System32\calc.exe
Process	reg.exe
Timestamp	2026-06-30 13:54:49 UTC

Analysis

Following the PowerShell execution alert, Wazuh detected a Sysmon Event ID 13 indicating that a registry Run Key had been created.

The investigation confirmed that reg.exe modified the following registry location:
HKCU\SOFTWARE\MICROSOFT\WINDOWS\CURRENTVERSION\RUN

A new value named Updater was configured to execute:
C:\WINDOWS\SYSTEM32\CALC.EXE

Although calc.exe is a legitimate Windows application, it was intentionally used in this lab to simulate malware persistence.

Registry Run Keys are a common persistence mechanism because Windows automatically executes these entries whenever the user logs on.

The chronological sequence between the PowerShell execution and the registry modification strongly suggests that persistence was intentionally established after gaining execution on the endpoint.

Indicators of Compromise (IOCs)

- Registry Run Key
- HKCU\Software\Microsoft\Windows\CurrentVersion\Run
- Value Name: Updater
- reg.exe
- Sysmon Event ID 13

5.3 Privilege Escalation Investigation

Initial Findings

Field	Value
User	SOC\testuser
Group	Administrators
Event ID	4732
Timestamp	2026-06-30 13:54:58 UTC

Analysis

The final stage of the investigation focused on the privilege escalation activity detected by Windows Security auditing.

Event ID 4732 confirmed that the account SOC\testuser was added to the local Administrators group.

Granting membership to the Administrators group significantly increases the privileges available to the compromised account, allowing the attacker to perform administrative operations such as disabling security controls, modifying system configurations, creating new privileged accounts, or deploying additional malware.

This activity occurred only seconds after the registry persistence event, demonstrating a clear attack progression:

- 1.Code execution
- 2.Persistence establishment
- 3.Privilege escalation

The investigation found no evidence of additional privileged accounts being modified during this incident.

Indicators of Compromise (IOCs)

- Security Event ID 4732
- Administrators Group
- User SOC\testuser
- Local group membership modification

6. Response

6.1 PowerShell Response

After confirming the suspicious PowerShell execution, immediate containment actions were initiated.

The following actions were performed:

- Verified running PowerShell processes.

```
PS C:\Windows\system32> Get-Process powershell

Handles NPM(K) PM(K) WS(K) CPU(s) Id SI ProcessName
-----
649 34 66284 79692 2.05 6544 1 powershell

PS C:\Windows\system32> Get-CimInstance Win32_Process | Where-Object {$_.Name -eq "powershell.exe"} | Select-Object ProcessId, CommandLine
ProcessId CommandLine
-----
6544 "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
```

Screenshot – Running PowerShell process verification

- > Confirmed that no additional malicious PowerShell instances were active.

- Verified that Microsoft Defender reported no active threats after execution.

```
PS C:\Windows\system32> Get-MpThreat
PS C:\Windows\system32>
```

Screenshot – Microsoft Defender status

- > No additional malicious PowerShell activity remained active after verification

6.2 Registry Persistence Response

Response Actions:

The malicious Registry Run Key persistence mechanism was successfully removed after confirming the unauthorized modification.

The following response actions were performed:

- Deleted the malicious Run Key (Updater).
- Verified that the Registry value had been successfully removed.
- Reviewed all remaining Run Keys for unauthorized persistence.
- Confirmed that no additional malicious persistence mechanisms were present.

```
C:\Windows\system32>reg delete "HKCU\Software\Microsoft\Windows\CurrentVersion\Run" /v Updater /f
The operation completed successfully.

C:\Windows\system32>reg query "HKCU\Software\Microsoft\Windows\CurrentVersion\Run"

HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Run
OneDrive REG_SZ "C:\Users\testuser\AppData\Local\Microsoft\OneDrive\OneDrive.exe" /background
Test123 REG_SZ C:\Windows\System32\calc.exe

C:\Windows\system32>reg delete "HKCU\Software\Microsoft\Windows\CurrentVersion\Run" /v Test123 /f
The operation completed successfully.

C:\Windows\system32>reg query "HKCU\Software\Microsoft\Windows\CurrentVersion\Run"

HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Run
OneDrive REG_SZ "C:\Users\testuser\AppData\Local\Microsoft\OneDrive\OneDrive.exe" /background
```

Screenshot – Registry value deletion & verification after deletion

- > The unauthorized Registry Run Key was successfully removed, and no additional persistence artifacts were identified during verification.

6.3 Privilege Escalation Response

Response Actions:

Following confirmation that the compromised account had been added to the local Administrators group, corrective actions were immediately performed.

The following actions were taken:

- Removed testuser from the Administrators group.
- Verified current membership of the Administrators group.
- Confirmed that only authorized administrative accounts remained.

```
PS C:\Windows\system32> Remove-LocalGroupMember -Group "Administrators" -Member "testuser"
PS C:\Windows\system32> Get-LocalGroupMember -Group "Administrators"

ObjectClass Name                PrincipalSource
-----
User          CLIENT\Administrator Local
User          CLIENT\Client        Local
Group         SOC\Domain Admins   ActiveDirectory
```

Screenshot – Removal of testuser & Verification of Administrators group membership

> Unauthorized administrative privileges were successfully revoked.

7. Recovery

Recovery Actions

Following the containment and remediation phases, several verification steps were performed to ensure that the affected endpoint had returned to a secure state.

- Verified that the malicious Registry Run Key had been successfully removed.
- Confirmed that the unauthorized Administrators group membership had been revoked.
- Verified that no suspicious PowerShell processes were running.
- Confirmed that no persistence mechanisms remained on the endpoint.
- Performed a Microsoft Defender scan to verify that no additional threats were present.
- Verified normal endpoint functionality after remediation.

Recovery Status

- Endpoint integrity verified.
- Persistence successfully removed.
- Privilege escalation reversed.
- No additional malicious activity observed.

8. Lessons Learned

Recovery Actions

Detection

- Sysmon provided detailed endpoint telemetry that significantly improved visibility into attacker activity.
- Custom Wazuh detection rules successfully identified suspicious PowerShell execution.
- Built-in Wazuh rules effectively detected Registry persistence and privilege escalation events.
- Correlating multiple alerts enabled identification of the complete attack chain.

Investigation

- Combining Windows Security Events with Sysmon logs greatly improved investigation efficiency.
- Mapping alerts to the MITRE ATT&CK framework simplified attack classification.
- Collecting evidence during every investigation phase facilitated incident documentation.

Response

- Early containment prevented the persistence mechanism from remaining active.
- Registry modifications should always be verified after remediation.
- Administrative group membership must be validated following any privilege escalation event.
- Verification activities are essential before declaring an incident resolved.

9. Recommendations

Recovery Actions

PowerShell

- Enable PowerShell Script Block Logging.
- Monitor suspicious PowerShell command-line arguments.
- Restrict PowerShell execution where operationally feasible.
- Alert on PowerShell execution using bypass parameters.

Registry Persistence

- Continuously monitor Registry Run Key modifications.
- Enable Sysmon Event ID 13 logging.
- Review Registry autorun entries during routine endpoint inspections.

Privilege Escalation

- Monitor Windows Security Event ID 4732.
- Review privileged group membership on a regular basis.
- Apply the Principle of Least Privilege.
- Alert on unexpected administrative group changes.

Monitoring

- Continue correlating Windows Security Events with Sysmon telemetry.
- Periodically validate custom detection rules.
- Perform regular incident response exercises to validate detection and response capabilities.

10. Conclusion

This incident simulated a realistic attack chain consisting of malicious PowerShell execution, Registry Run Key persistence, and local privilege escalation.

Throughout the exercise, Wazuh successfully detected each attack phase using a combination of custom and built-in detection rules. The investigation confirmed the sequence of attacker activities, while the response process successfully removed the persistence mechanism, revoked unauthorized administrative privileges, and verified that no malicious PowerShell activity remained on the endpoint.

Following recovery verification, the affected endpoint was confirmed to be operating normally with no remaining indicators of compromise.

Incident Status: Closed